Stress Concentrations

ENGI 2221 Coursework Report

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October 23, 2026

Abstract

This report analyses stress concentrations in an aircraft fuselage window and a notched strip. Analytical and finite element methods are used to determine these stresses. The latter study of the notched strip focuses on the relationships between different geometric parameters and the resulting stress concentration factors.

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1 Introduction

This report examines stress concentrations in an aircraft fuselage window and in a strip with opposing U-shaped notches. The fuselages is modelled as a thin-walled pressure vessel, this is done to determine axial and hoop stresses. The far-field hoop stress is applied to the elliptical opening to evaluate the resulting stress distribution and assess the analytical assumptions made. Computational simulations are subsequently used to investigate how the stress concentration factor varies with the strips geometry.

Table 1: Fuselage variables

Sym.	Description	Unit	Value
W	Fuselage width	m	3.5
t	Wall thickness	m	0.001
p_{int}	Internal pressure	KPa	75.29
p_{ext}	External pressure	KPa	30.06
Δp	Pressure difference	KPa	45.23
r	Fuselage radius	m	1.75
a	Semi-major axis	m	0.125
b	Semi-minor axis	m	0.11875
$\sigma_{\theta\theta}$	Hoop stress	MPa	79.15
σ_{zz}	Axial stress	MPa	39.58
α_0	Parameter	rad	1.8318
β	Angle around window	degrees	$[0, 360]$
$\sigma_{\beta\beta}$	Tangential stress	MPa	see Fig. 2
K_t	Stress concentration factor	–	see results
$K_{t,\text{theory}}$	Theoretical K_t	–	3.105

Table 2: Notched Strip variables

Sym.	Description	Unit	Value
H	Strip width	mm	15 – 65
d	Notch separation	mm	8
r	Notch radius	mm	0.4 – 2.4
L	Strip length	mm	10 – 120
S	Applied traction	MPa	100
$\sigma_{\max}$	Max principal stress	MPa	see results
σ_{nom}	Nominal stress	MPa	100
K_t	Stress concentration factor	–	see results
H/d	Width ratio	–	1.9 – 8.1
r/d	Radius ratio	–	10^{-2} – 0.3
L/H	Length ratio	–	0.25 – 3.0

2 Theory

This section presents the equations and assumptions used in the report.

2.1 Stress Concentration Factor Equations

The stress concentration factor, K_t, is defined as the ratio of the maximum stress, $\sigma_{\max}$, at a discontinuity to the nominal applied stress, σ_{nom}:

$$K_t = \frac{\sigma_{\max}}{\sigma_{\text{nom}}} \tag{1}$$

For the case of an elliptical hole in a large plate subjected to uniaxial tension, the stress concentration factor can be expressed as:

$$K_t = 1 + \frac{2a}{b} \tag{2}$$

This expression shows that the stress concentration increases as the window elongates (a/b increases). This formula is Equation (4.57) reported in [1].

2.2 Thin-Walled Pressure Vessel Assumptions

The aircraft fuselage is modelled as a closed-ended thin-walled cylindrical pressure vessel [2]. The closed-ended assumption is valid due to the nose and tail of the aeroplane transmitting axial loads to the cylinder walls. The thin-walled assumption is valid due to the fuselage radius being far larger than the wall thickness ($r/t = 1750 \gg 1$); and therefore the stress variation through the wall thickness is negligble.
The hoop (circumferential) stress in a thin-walled cylindrical pressure vessel is given by equation 3 [3, p. 399]:

$$\sigma_{\theta\theta} = \frac{\Delta p\, r}{t} \tag{3}$$

where Δp is the pressure difference across the wall thickness, t, and r is the fuselage radius.

The axial stress in a thin-walled cylindrical pressure vessel is given by equation 4 [2].

$$\sigma_{zz} = \frac{\Delta p\, r}{2t} \tag{4}$$

These stresses are the far-field loading stresses applied to the elliptical opening in the following stress concentration analysis.

2.3 Plane Stress Assumption

In this analysis, plane stress is assumed; the stress component acting through the thickness of the fuselage wall is also assumed to be zero ($\sigma_{nn} = 0$). This assumption can be taken as the thickness of the wall is small relative to the cylinder radius ($r/t = 1750$) so the normal stress is negligible and plane stress applies.

The tangential stress around any point on the edge of the elliptical hole is given by equation 5 (see Ref. [1], p. 252, Eq. (4.61)).

It is assumed that the fuselage is an infinite plate containing a single elliptical hole under uniaxial tension applied parallel to the major axis of the ellipse. The plate is assumed to be large so that any edge or boundary stress can be neglected, and sufficiently thin such that plane stress conditions apply. These assumptions are used in finding the equation for stress around elliptical holes developed by Inglis [4].

$$\sigma_{\beta\beta}(\beta) = S\,\frac{\sinh(2\alpha_0) + 1 - e^{2\alpha_0}\cos(2\beta)}{\cosh(2\alpha_0) - \cos(2\beta)} \tag{5}$$

where $\beta \in [0°, 360°]$, S is the nominal far-field stress and

$$\alpha_0 = \tanh^{-1}\left(\frac{b}{a}\right)$$

The geometry and coordinate system used to describe this stress distribution are shown in Figure 1.

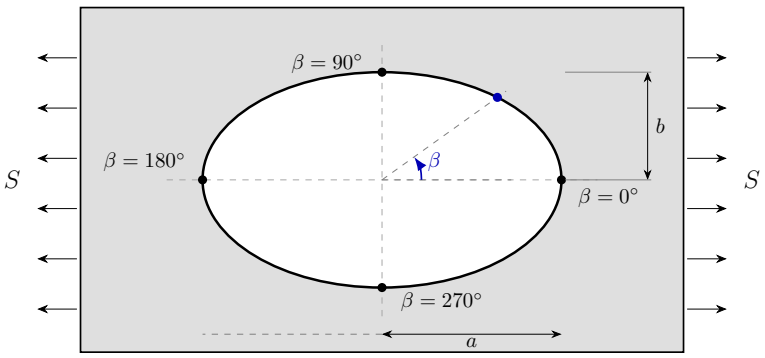


Figure 1: Stress S applied parallel to the major axis of the elliptical hole. The angular coordinate β is measured counter-clockwise from $\beta = 0°$.

3 Theoretical Analysis of Fuselage and Window Opening Stresses

This section covers the stresses in a cylindrical aircraft fuselage and the stress distribution around an elliptical window in the fuselage.

A typical cruising altitude of 9,144 m (30,000 ft) was adopted for this analysis, at which the atmospheric pressure is $30.06\,\text{kPa}$ [5] [6]. Federal Aviation Regulations require the cabin to maintain an internal cabin pressure of no less than the atmospheric equivalent of 2,438 m, corresponding to a cabin pressure of $75.29\,\text{kPa}$ [7]. Therefore, this analysis will use an internal cabin pressure of $75.29\,\text{kPa}$, and an atmospheric pressure of $30.06\,\text{kPa}$.

The pressure acting on the fuselage wall is caused by the difference between the internal cabin pressure and the and external atmospheric pressure.

$$\Delta p = p_{\text{int}} - p_{\text{ext}} = 75.29 - 30.06 = 45.23\,\text{kPa} \tag{6}$$

This pressure differential produces stresses in the fuselage wall, modelled using the assumptions developed in subsection 2.2. The resulting hoop and axial stresses of the windowless fuselage are:

$$\sigma_{\theta\theta} = \frac{\Delta p\, r}{t} = 79.15\,\text{MPa} \tag{7}$$

$$\sigma_{zz} = \frac{\Delta p\, r}{2t} = 39.58\text{MPa} \tag{8}$$

Since the major axis is aligned with the hoop direction, the hoop stress is taken as the nominal far-field stress S. The tangential stress distribution is shown in Figure 2.

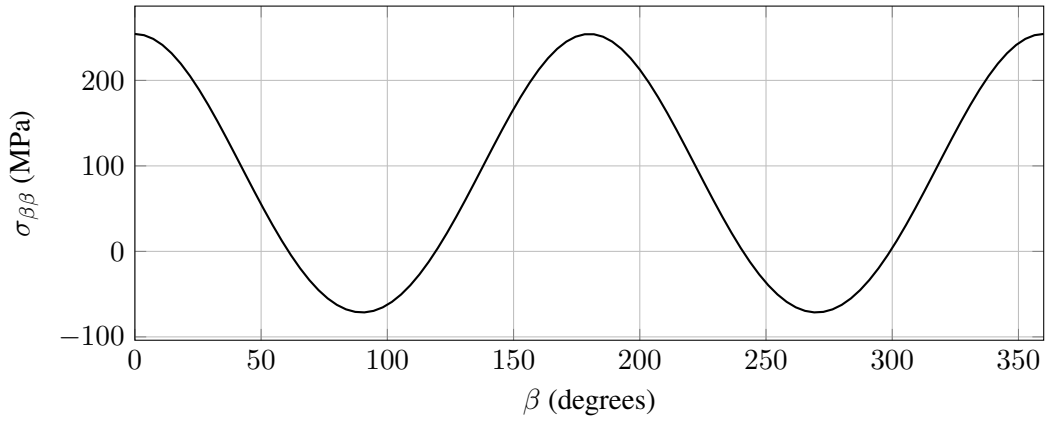


Figure 2: Tangential stress distribution around the elliptical window opening.

Figure 2 illustrates that with an elliptical shaped opening there are significant variations in stress around the opening. This leads to aircraft fuselage design optimising the shape and reinforcing these openings to reduce the effect of high stress cycling across the top and bottom of the elipse (most stressed points) as the plane increases and decreases its altitude.

3.1 Comparison with Material Yield Strength

The maximum tangential stress occurs at the ends of the major axis of the window, where the stress concentration is greatest. Using the theoretical stress concentration factor,

$$K_{t,\text{theory}} = 1 + \frac{2a}{b} = 3.105, \tag{9}$$

and the hoop stress as the nominal far-field stress,

$$\sigma_{\max} = K_t\, \sigma_{\theta\theta} = 3.105 \times 79.15 = 246 \text{ MPa}. \tag{10}$$

Typical fuselage alloys such as AA 2024-T3 have yield stresses of about 345 MPa [8]. Since the predicted maximum stress does not exceed this value, yielding is not expected at the elliptical opening. However, aeroplane fuselages are designed with factors of safety, therefore reinforcements are used to reduce local stress concentrations around the opening.

3.2 Assumptions and Limitations

The fuselage is modelled as a thin-walled closed-ended cylinder ($r/t \gg 1$), allowing the use of thin cylinders equations 3 and 4, and neglecting stress variation through the fuselage wall thickness.

Additionally, the axial stress (σ_{zz}) is assumed to be uniform throughout the fuselage, and plane stress conditions are assumed ($\sigma_{nn} = 0$).

The fuselage wall is idealised as an infinite flat plate containing a single elliptical hole, neglecting that it is not flat but in fact curved. In addition, equation 5 assumes uniaxial loading, taking the hoop stress $\sigma_{\theta\theta}$ as the only far-field stress. In reality, the fuselage is subjected to axial stress aswell as hoop stress (a biaxial stress state), with $\sigma_{zz} = \sigma_{\theta\theta}/2$ (from subsection 2.2).

The formula for the tangential stress around an elliptical opening under biaxial loading is given by [1] as

$$\sigma_{\beta\beta} = \frac{(\sigma_1 + \sigma_2)\sinh 2\alpha_0 + (\sigma_2 - \sigma_1)\left(1 - e^{2\alpha_0}\cos 2\beta\right)}{\cosh 2\alpha_0 - \cos 2\beta} \tag{11}$$

where $\sigma_1 = \sigma_{\theta\theta}$ and $\sigma_2 = \sigma_{zz}$.

This equation reveals that the opening has a maximum stress of $\sigma_{\max} \approx 206$ MPa compared to 246 MPa for the uniaxial case. The uniaxial equation therefore overestimates the maximum stress by approximately 16%.

The opening is assumed to be a perfect ellipse in a homogeneous, and linearly elastic material. Real elipses are not included in the fuselage due to manufacturing constraints. Corner radii replacing any sharp edge (such as the edge of an elipse), due to manufacturing constraints, has been shown to reduce the local concentration factor by up to 50% [9]. This suggests that neglecting manufacturing imperfections and plastic regions limits the accuracy of solutions and overestimates maximum stresses when applied to real fuselages, which may lead to inefficient design of fuselages.

3.3 Effect of Multiple Window Openings

The prior analysis investigates a single window opening; however, real fuselages contain multiple openings. When analysing multiple openings close together, the validity of the large plate assumption is reduced.

For a row of openings, Peterson shows that the interference effect is:

$$\frac{K_t}{K_{t0}} \propto \frac{a^2}{c} \tag{12}$$

where K_t is the stress concentration factor for multiple openings, K_{t0} is the stress concentration factor for a single isolated opening, a is the semi-major axis of the opening, and c is the centre-to-centre spacing [1].

This equation suggests that increasing the size of the openings or reducing their spacing increases the stress concentration factor compared to a single opening. Furthermore, windows that are placed close together can produce higher stresses than those found in the prior single-opening analysis, with the highest stress occurring on the final openings in the row.

4 Computational Simulation of Stress Concentrations

This section presents a computational analysis of stress concentrations in a strip containing two U-shaped notches subjected to a far-field stress, S, of 100 MPa. A diagram of the strip's geometry and its defining parameters is shown in Figure 3, and described in Table 2. A diagram of an arbitrary contour plot is also seen in Figure 3

The stress concentration factor, K_t, defined in Eq. 1, was investigated at the notch for varying values of the strip's geometry H/d, r/d, and L. In each analysis, one parameter was varied and the rest kept constant in order to understand the individual effect of the parameter on the stress concentration.

The resulting variations in K_t are presented in the following subsections.

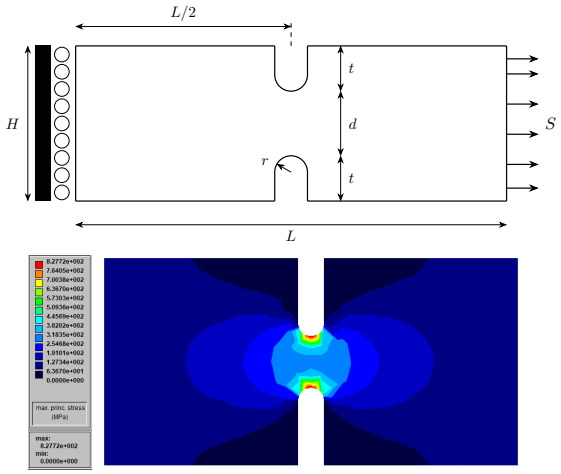


Figure 3: Geometry of the notched strip together with the stress legend and Concept Analyst contour plot.

4.1 Analysis of strip width H/d

Figure 4 shows the variation of K_t with H/d, where K_t is the stress concentration factor and H/d is the strip width relative to the notch separation. K_t varies approximately linearly with H/d over the range used in this analysis.

A fixed value of d gives an increasing width of material aound the notches as H/d increases. This allows greater stress concentrations around the notches. The linearity of the graph susggests that the strip is still elastic and has not yielded.

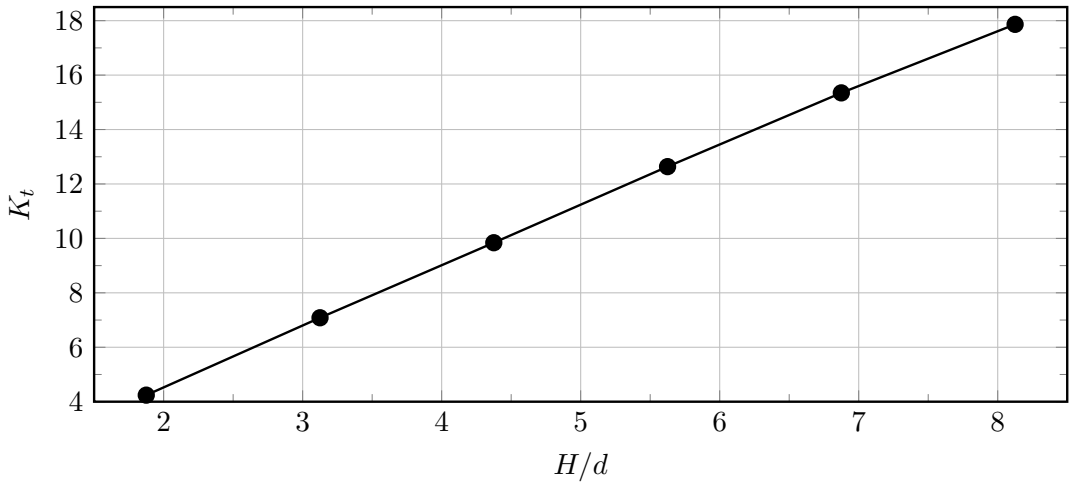


Figure 4: Variation of stress concentration factor K_t with width ratio H/d for a strip under 100 MPa traction.

4.2 Analysis of notch radius r/d

Figure 5 shows how the stress concentration factor K_t varies with the ratio r/d, where r is the radius of the notch.

As r/d increases, K_t decreases. Increasing r/d corresponds to increasing the notch radius r, as d is constant. This reduces the stress localisation in the area between the notches and around the notches. The notches are sharp and generate high stresses when r/d is small, this results in large values of K_t. Stresses reduce as r/d increases, this is due to more blunt notches and greater area around and between the notches. The change in gradient $\frac{dK_t}{d(r/d)}$ becomes more shallow as r/d increases, leading to K_t reaching a final constant value as r/d increases further.

This analysis is in agreement with stress concentration theory, as sharper notches are suggested to produce higher stress concentrations [1].

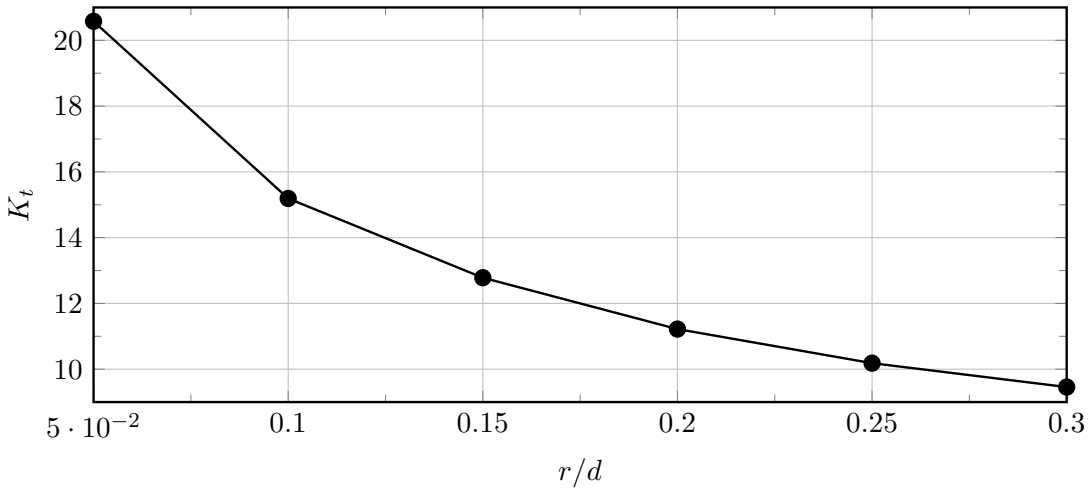


Figure 5: Variation of stress concentration factor K_t with notch radius ratio r/d for a strip under 100 MPa traction.

4.3 Analysis of Strip Length L/H

Figure 6 shows the variation of K_t with the ratio L/H, where L is the strip length and H is the width of the strip.

A significant increase in K_t is seen for minor values of L/H, and very large stress concentrations are observed when the strip is short relative to its width. When the length is small relative to its width, the distance from the stressed edges has a significant influence on the stresses observed in the strip, and therefore the assumption of an effectively infinite plate is not valid for these values of L/H.

An increase in L/H leads to a decrease in K_t, this then leads to an asymptote at a final value of K_t. For any values of $L/H \gtrsim 1.5$, an increase in length produces no measurable change in K_t. This suggests that the stresses become independent of the length-width ratio at a value of $L/H \gtrsim 1.5$, and therefore this threshold is where the infinite plate assumption becomes valid.

This finding is consistent with results for short flat bars with opposing U-shaped notches, where it has been shown that classic stress concentration data is not valid for small values of L/H [10].

These results demonstrate that the length of the member is a critical parameter in analysing the stress concentration factor. And applying infinite plate assumptions to short members can result in significant underestimation of stress concentration factors; such as in the prior analysis of the elliptical fuselage opening given other discontinuities were to be located near the window being investigated.

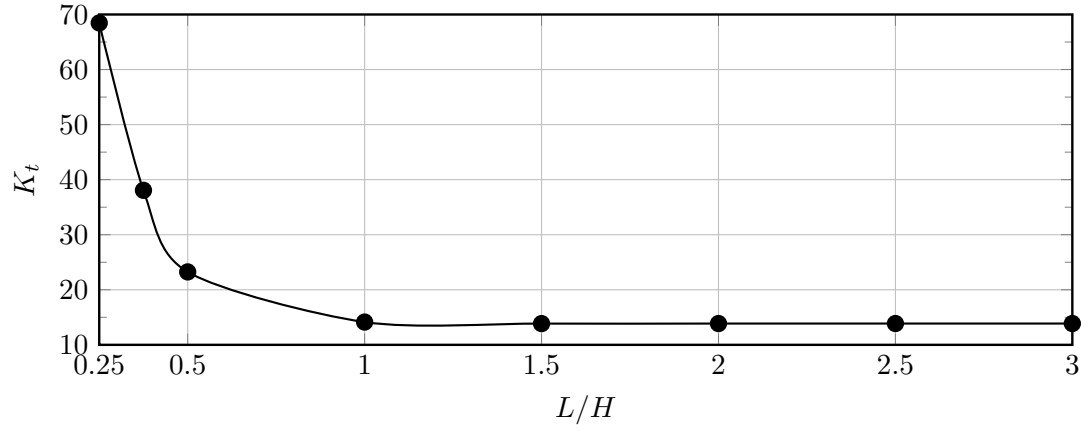


Figure 6: Variation of stress concentration factor K_t with length ratio L/H for fixed parameter $H/d = 5$ and $r/d = 0.125$ under 100 MPa traction.

5 Conclusion

This report analysed stress concentrations in aeroplane fuselage windows, and around notches in strips. The section which analysed an eliptical opening found $K_t \approx 3.1$, which suggested that discontinuities increase local stresses significantly, and it also highlighted that most assumptions taken in this analysis make the final result inaccurate, such as the uniaxial loading assumption.

The computational study of the strip revealed that the stress concentration factor increases with H/d, decreses with r/d as the notch becomes less sharp, and becomes indepenent of L/H once L/H is greater than 1.5.

In summary, the analysis of stress concentrations must take into account geometry, ratios of dimensions, and key assumptions when required to find accurate results.

References

[1] Walter D. Pilkey, Deborah F. Pilkey, and Zhuming Bi. *Peterson's Stress Concentration Factors*. John Wiley & Sons, 4 edition, 2020.

[2] David Roylance. Pressure vessels. Course notes, Massachusetts Institute of Technology, 2001.

[3] Eric R. Johnson. *Aerospace Structures*. Virginia Tech Publishing, Blacksburg, VA, 2022.

[4] C. E. Inglis. Stresses in a plate due to the presence of cracks and sharp corners. *Transactions of the Institution of Naval Architects*, 55:219–230, 1913.

[5] Aerospace Medical Association, Aviation Safety Committee, Civil Aviation Subcommittee. Cabin cruising altitudes for regular transport aircraft. *Aviation, Space, and Environmental Medicine*, 79(4):433–439, 2008.

[6] Jean-Luc Boiffier and Laurent Bovet. Why commercial jet planes cruise at 30,000 ft? conceptual optimal cruise altitude. In *Proceedings of the 27th Congress of the International Council of the Aeronautical Sciences*, Nice, France, 2010. ICAS 2010, 19–24 September.

[7] Federal Aviation Administration. 14 CFR 25.841 – pressurized cabins. Electronic Code of Federal Regulations (eCFR), 2026. Accessed 2026-04-13.

[8] M. Cabrini, S. Bocchi, G. D'Urso, C. Giardini, S. Lorenzi, C. Testa, and T. Pastore. Stress corrosion cracking of friction stir-welded aa-2024 t3 alloy. *Materials*, 13(11), 2020.

[9] Nikola Momcilovic, Milorad Motok, and Tasko Maneski. Stress concentration on the contour of a plate opening: Analytical, numerical and experimental approach. *Journal of Theoretical and Applied Mechanics*, 51(4):1003–1012, 2013.

[10] C. J. Gomes, N. Troyani, C. Morillo, S. Gregory, V. Gerardo, and Y. Pollonais. Theoretical stress concentration factors for short flat tension bars with opposite u-shaped notches. *The Journal of Strain Analysis for Engineering Design*, 40(4):345–355, 2005.

A Appendix

Study 1: H/d

H	Depth	H/d	$\sigma_{\max}$	K_t
15	3.5	1.875	424	4.24
25	8.5	3.125	708	7.08
35	13.5	4.375	984	9.84
45	18.5	5.625	1264	12.64
55	23.5	6.875	1535	15.35
65	28.5	8.125	1787	17.87
100	46	12.5	2685	26.85
160	76	20	4355	43.55

Study 2: r/d

r/d	r	$\sigma_{\max}$	K_t
0.05	0.4	2058	20.58
0.10	0.8	1519	15.19
0.15	1.2	1278	12.78
0.20	1.6	1122	11.22
0.25	2.0	1018	10.18
0.30	2.4	946	9.46

Study 3: L/H

L/H	L	$\sigma_{\max}$	K_t
0.25	10	6847	68.47
0.375	15	3807	38.07
0.5	20	2325	23.25
1.0	40	1414	14.14
1.5	60	1386	13.86
2.0	80	1387	13.87
2.5	100	1387	13.87
3.0	120	1387	13.87

Table 3: Summary of simulation results for the three analysis. All lengths are in mm and stresses are in MPa.